Anaphora Resolution

You may use any programming language to do this assignment. Please submit the following documents:

1. A PDF report answering the questions mentioned below
2. Your code and README file in a tarball archive

Reference: Chapter 21 from Jurafsky and Martin 2nd edition (J&M henceforth), available here.

Dataset: Download the GAP (Gender-Balanced Coreference Data) dataset from here.

Task: Identify the anaphor of each pronoun in the dataset using your own implementation of a simple Logistic Regression classifier. Then submit a report after performing the following operations on the dataset you downloaded.

Problem 1

(10 marks) For each discourse segment in the test set, extract all the candidate noun phrases that can potentially serve as the antecedent of the given pronoun. You need to use a standard POS tagger.

Problem 2

(5 marks) Randomly assign an antecedent to the pronoun and calculate average accuracy for 100 random assignment runs.

Problem 3

(5 marks) Select the most recent antecedent of the pronoun and compute accuracy.

Problem 4

(10 marks) Implement a Logistic Regression classifier by extracting features from the candidate noun phrases (also from the parse trees from the Hobbs algorithm implementation below) you extracted and report precision, recall and F-score. Feature weights should be learned from the training data, and the evaluation needs to be done on the test set. Hyperparameter tuning should be done on the development set.

Problem 5

(5 marks) Print out the features and their weights learned by the LR model. Write a short note on the top-5 features (in terms of their weight) and how they can be interpreted.

Problem 6

(5 marks) Write a report describing the errors made by your implementation and how the algorithm can be improved.

Problem 7

(10 marks) Implement the rule-based Hobbs' algorithm (use a standard parser to get Context Free Grammar trees) and report it's performance (in terms of F-scores).

Problem 8

(5 marks) Compare the results of the LR algorithm with the Hobbs algorithm both qualitatively and quantitatively.

Extra Credit

(10 marks) Use a standard toolkit like scikit-learn to implement Logistic Regression and answer all the questions above.